

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 24fa0d67 · Data hash 0e67...9ce8

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 15:54 and 15:54 UTC. Reported results: final_train_accuracy 100, final_val_accuracy 97.5. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	24fa0d67-4ada-4055-bb24-d0bbf7257b07
Initialized	2026-10-07T15:54:12Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	a6f61333806c5b85596eceb8ef35e371eae9d369608e78e5a9ceb8de8cc37ae9
Disclosure	open
Source code, run 1	c36ca4390d911cbeaea8379f37911707a49acea0eb89577eac7b587f258d062b

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	28.5 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
train_loss	0.083	600 (13 readings)	explicit
val_loss	0.074	600 (13 readings)	explicit
train_accuracy	100	5 (5 readings)	explicit
val_accuracy	97.5	5 (5 readings)	explicit

Metric	Final value	Step	Source
training_minutes	0.32		explicit

Inline claim	Value	Phase	Output line
final_train_accuracy	100		39
final_val_accuracy	97.5		40

4 Execution evidence

Coherence

Coherent (0.49): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	279	30.1 s	2.90 GB	98%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 256 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	15:54:12	a92737e1ac275733	256
2	run_end	15:54:40	a92737e1ac275733	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 260 files read, 5 files written, 3 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T15:54:41.642621+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	0e679d2750be501506cdce3c313cf4c7517b8764e6495f6425cfc40170929ce8
Code bundle hash	c48ba9b0e18a012a4ec84ffd175e42d62e13f1ecf9979d7a4a2a5631760909c4

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
train_loss	2.153	0	2	explicit
val_loss	2.392	0	3	explicit
train_loss	0.617	50	4	explicit
val_loss	0.637	50	5	explicit
train_loss	0.523	100	6	explicit
val_loss	0.557	100	7	explicit
train_accuracy	70	1	8	explicit
val_accuracy	72.5	1	9	explicit
train_loss	0.561	150	10	explicit
val_loss	0.489	150	11	explicit
train_loss	0.419	200	12	explicit
val_loss	0.397	200	13	explicit
train_loss	0.409	250	14	explicit
val_loss	0.353	250	15	explicit
train_accuracy	82.5	2	16	explicit
val_accuracy	85	2	17	explicit
train_loss	0.333	300	18	explicit
val_loss	0.32	300	19	explicit
train_loss	0.34	350	20	explicit
val_loss	0.306	350	21	explicit
train_accuracy	90	3	22	explicit
val_accuracy	90	3	23	explicit
train_loss	0.136	400	24	explicit
val_loss	0.2	400	25	explicit
train_loss	0.153	450	26	explicit
val_loss	0.132	450	27	explicit
train_loss	0.222	500	28	explicit
val_loss	0.137	500	29	explicit
train_accuracy	100	4	30	explicit
val_accuracy	97.5	4	31	explicit
train_loss	0.207	550	32	explicit
val_loss	0.143	550	33	explicit
train_loss	0.083	600	34	explicit
val_loss	0.074	600	35	explicit
train_accuracy	100	5	36	explicit
val_accuracy	97.5	5	37	explicit
training_minutes	0.32		38	explicit

Phase	Line	Time	CPU time	Memory	GPU
finetune	1	15:54:12	1.3 s	0.10 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	15:54:12	a92737e1ac275733	256
1	phase finetune	15:54:12	a92737e1ac275733	0
2	run_end	15:54:40	a92737e1ac275733	0

Event	Path or command	Hash
read	kv_run.sh	
read	ch06/01_main-chapter-code/gpt_class_finetune.py	
read	.github/ISSUE_TEMPLATE/bug-report.yaml	
read	.github/scripts/check_double_quotes.py	
read	.github/scripts/check_links.py	
read	.github/scripts/test_check_links.py	
read	.github/workflows/basic-tests-latest-python.yml	
read	.github/workflows/basic-tests-linux-uv.yml	
read	.github/workflows/basic-tests-macos-uv.yml	
read	.github/workflows/basic-tests-old-pytorch.yml	
read	.github/workflows/basic-tests-pip.yml	
read	.github/workflows/basic-tests-pixi.yml	
read	.github/workflows/basic-tests-pytorch-rc.yml	
read	.github/workflows/basic-tests-windows-uv-pip.yml	
read	.github/workflows/check-links.yml	
read	.github/workflows/check-spelling-errors.yml	
read	.github/workflows/pep8-linter.yml	
read	.github/workflows/required-checks.yml	
read	.github/workflows/retry-required-checks.yml	
read	appendix-A/01_main-chapter-code/DDP-script-torchrun.py	
read	appendix-A/01_main-chapter-code/DDP-script.py	
read	appendix-A/01_main-chapter-code/code-part1.ipynb	
read	appendix-A/01_main-chapter-code/code-part2.ipynb	
read	appendix-A/01_main-chapter-code/exercise-solutions.ipynb	
read	appendix-D/01_main-chapter-code/appendix-D.ipynb	
read	appendix-D/01_main-chapter-code/previous_chapters.py	
read	appendix-E/01_main-chapter-code/appendix-E.ipynb	
read	appendix-E/01_main-chapter-code/gpt_download.py	
read	appendix-E/01_main-chapter-code/previous_chapters.py	
read	ch02/01_main-chapter-code/ch02.ipynb	
read	ch02/01_main-chapter-code/dataloader.ipynb	
read	ch02/01_main-chapter-code/exercise-solutions.ipynb	
read	ch02/02_bonus_bytepair-encoder/bpe_openai_gpt2.py	
read	ch02/02_bonus_bytepair-encoder/compare-bpe-tiktoken.ipynb	
read	ch02/02_bonus_bytepair-encoder/gpt2_model/encoder.json	
read	ch02/03_bonus_embedding-vs-matmul/embeddings-and-linear-layers.ipynb	
read	ch02/04_bonus_dataloader-intuition/dataloader-intuition.ipynb	
read	ch02/05_bpe-from-scratch/bpe-from-scratch-simple.ipynb	
read	ch02/05_bpe-from-scratch/bpe-from-scratch.ipynb	
read	ch02/05_bpe-from-scratch/tests.py	
read	ch02/06_bonus_simple-tokenizer-v3/simple-tokenizer-v3.ipynb	

Event	Path or command	Hash
read	ch03/01_main-chapter-code/ch03.ipynb	
read	ch03/01_main-chapter-code/exercise-solutions.ipynb	
read	ch03/01_main-chapter-code/multihead-attention.ipynb	
read	ch03/02_bonus_efficient-multihead-attention/mha-implementations.ipynb	
read	ch03/02_bonus_efficient-multihead-attention/tests/test_mha_implementations...	
read	ch03/03_understanding-buffers/understanding-buffers.ipynb	
read	ch04/01_main-chapter-code/ch04.ipynb	
read	ch04/01_main-chapter-code/exercise-solutions.ipynb	
read	ch04/01_main-chapter-code/gpt.py	
read	ch04/01_main-chapter-code/previous_chapters.py	
read	ch04/01_main-chapter-code/tests.py	
read	ch04/02_performance-analysis/flops-analysis.ipynb	
read	ch04/03_kv-cache/gpt_ch04.py	
read	ch04/03_kv-cache/gpt_with_kv_cache.py	
read	ch04/03_kv-cache/gpt_with_kv_cache_optimized.py	
read	ch04/03_kv-cache/tests.py	
read	ch04/04_gqa/gpt_with_kv_gqa.py	
read	ch04/04_gqa/gpt_with_kv_mha.py	
read	ch04/04_gqa/memory_estimator_gqa.py	
read	ch04/04_gqa/plot_memory_estimates_gqa.py	
read	ch04/05_mla/gpt_with_kv_mha.py	
read	ch04/05_mla/gpt_with_kv_mla.py	
read	ch04/05_mla/memory_estimator_mla.py	
read	ch04/05_mla/plot_memory_estimates_mla.py	
read	ch04/06_swa/gpt_with_kv_mha.py	
read	ch04/06_swa/gpt_with_kv_swa.py	
read	ch04/06_swa/memory_estimator_swa.py	
read	ch04/06_swa/plot_memory_estimates_swa.py	
read	ch04/06_swa/tests.py	
read	ch04/07_moe/gpt_with_kv_ffn.py	
read	ch04/07_moe/gpt_with_kv_moe.py	
read	ch04/07_moe/memory_estimator_moe.py	
read	ch04/07_moe/plot_memory_estimates_moe.py	
read	ch04/08_deltanet/plot_memory_estimates_gated_deltanet.py	
read	ch04/09_dsa/gpt_with_kv_dsa.py	
read	ch04/09_dsa/test_dsa.py	
read	ch04/10_kv-sharing/gpt_with_kv_mha.py	
read	ch04/10_kv-sharing/gpt_with_kv_sharing.py	
read	ch04/10_kv-sharing/memory_estimator_kv_sharing.py	
read	ch04/10_kv-sharing/plot_memory_estimates_kv_sharing.py	
read	ch04/10_kv-sharing/tests.py	
read	ch05/01_main-chapter-code/ch05.ipynb	
read	ch05/01_main-chapter-code/exercise-solutions.ipynb	
read	ch05/01_main-chapter-code/gpt_download.py	
read	ch05/01_main-chapter-code/gpt_generate.py	
read	ch05/01_main-chapter-code/gpt_train.py	
read	ch05/01_main-chapter-code/previous_chapters.py	
read	ch05/01_main-chapter-code/tests.py	
read	ch05/02_alternative_weight_loading/weight-loading-hf-safetensors.ipynb	
read	ch05/02_alternative_weight_loading/weight-loading-hf-transformers.ipynb	

Event	Path or command	Hash
read	ch05/02_alternative_weight_loading/weight-loading-pytorch.ipynb	
read	ch05/03_bonus_pretraining_on_gutenberg/prepare_dataset.py	
read	ch05/03_bonus_pretraining_on_gutenberg/pretraining_simple.py	
read	ch05/03_bonus_pretraining_on_gutenberg/tests.py	
read	ch05/05_bonus_hparam_tuning/hparam_search.py	
read	ch05/06_user_interface/app_orig.py	
read	ch05/06_user_interface/app_own.py	
read	ch05/07_gpt_to_llama/config.json	
read	ch05/07_gpt_to_llama/convert-gpt-to-llama2.ipynb	
read	ch05/07_gpt_to_llama/convert-llama2-to-llama3.ipynb	
read	ch05/07_gpt_to_llama/previous_chapters.py	
read	ch05/07_gpt_to_llama/standalone-llama32.ipynb	
read	ch05/07_gpt_to_llama/tests/test_llama32_nb.py	
read	ch05/07_gpt_to_llama/tests/test_rope_and_parts.py	
read	ch05/08_memory_efficient_weight_loading/memory-efficient-state-dict.ipynb	
read	ch05/08_memory_efficient_weight_loading/previous_chapters.py	
read	ch05/09_extending-tokenizers/extend-tiktoken.ipynb	
read	ch05/10_llm-training-speed/00_orig.py	
read	ch05/10_llm-training-speed/01_opt_single_gpu.py	
read	ch05/10_llm-training-speed/02_opt_multi_gpu_ddp.py	
read	ch05/11_qwen3/qwen3-chat-interface/qwen3-chat-interface-multiturn.py	
read	ch05/11_qwen3/qwen3-chat-interface/qwen3-chat-interface.py	
read	ch05/11_qwen3/standalone-qwen3-moe-plus-kvcache.ipynb	
read	ch05/11_qwen3/standalone-qwen3-moe.ipynb	
read	ch05/11_qwen3/standalone-qwen3-plus-kvcache.ipynb	
read	ch05/11_qwen3/standalone-qwen3.ipynb	
read	ch05/11_qwen3/tests/test_qwen3_kvcache_nb.py	
read	ch05/11_qwen3/tests/test_qwen3_nb.py	
read	ch05/12_gemma3/standalone-gemma3-plus-kvcache.ipynb	
read	ch05/12_gemma3/standalone-gemma3.ipynb	
read	ch05/12_gemma3/tests/gemma3-transformers-ref.ipynb	
read	ch05/12_gemma3/tests/gemma3_layer_debugger.py	
read	ch05/12_gemma3/tests/gemma3_layer_debugger_detailed.py	
read	ch05/12_gemma3/tests/test_gemma3_kv_nb.py	
read	ch05/12_gemma3/tests/test_gemma3_nb.py	
read	ch05/13_olmo3/standalone-olmo3-plus-kv-cache.ipynb	
read	ch05/13_olmo3/standalone-olmo3.ipynb	
read	ch05/13_olmo3/tests/olmo3_layer_debugger.py	
read	ch05/13_olmo3/tests/test_olmo3_kvcache_nb.py	
read	ch05/13_olmo3/tests/test_olmo3_nb.py	
read	ch05/14_ch05_with_other_llms/ch05-llama32.ipynb	
read	ch05/14_ch05_with_other_llms/ch05-qwen3.ipynb	
read	ch05/15_tiny-aya/standalone-tiny-aya-plus-kv-cache.ipynb	
read	ch05/15_tiny-aya/standalone-tiny-aya.ipynb	
read	ch05/15_tiny-aya/tests/test_tiny_aya_kvcache_nb.py	
read	ch05/15_tiny-aya/tests/test_tiny_aya_nb.py	
read	ch05/15_tiny-aya/tests/tiny_aya_layer_debugger.py	
read	ch05/16_qwen3.5/qwen3.5-plus-kv-cache.ipynb	
read	ch05/16_qwen3.5/qwen3.5.ipynb	
read	ch05/16_qwen3.5/qwen3_5_transformers.py	

Event	Path or command	Hash
read	ch05/16_qwen3.5/tests/qwen3_5_layer_debugger.py	
read	ch05/16_qwen3.5/tests/test_qwen3_5_nb.py	
read	ch05/17_gemma4/standalone-gemma4-plus-kvcache.ipynb	
read	ch05/17_gemma4/standalone-gemma4.ipynb	
read	ch05/17_gemma4/tests/test_e2b/gemma4-e2b-transformers-ref.ipynb	
read	ch05/17_gemma4/tests/test_e2b/gemma4_e2b_layer_debugger.py	
read	ch05/17_gemma4/tests/test_e4b/gemma4-e4b-transformers-ref.ipynb	
read	ch05/17_gemma4/tests/test_e4b/gemma4_e4b_layer_debugger.py	
read	ch05/17_gemma4/tests/test_gemma4_nb.py	
read	ch05/18_muon/gpt_train.py	
read	ch05/18_muon/gpt_train_muon.py	
read	ch05/18_muon/previous_chapters.py	
read	ch05/19_cpu_mps_differences/1_cpu_mps_random_generator.ipynb	
read	ch06/01_main-chapter-code/ch06.ipynb	
read	ch06/01_main-chapter-code/exercise-solutions.ipynb	
read	ch06/01_main-chapter-code/gpt2/124M/encoder.json	
read	ch06/01_main-chapter-code/gpt2/124M/hparams.json	
read	ch06/01_main-chapter-code/gpt_download.py	
read	ch06/01_main-chapter-code/load-finetuned-model.ipynb	
read	ch06/01_main-chapter-code/previous_chapters.py	
read	ch06/01_main-chapter-code/tests.py	
read	ch06/02_bonus_additional-experiments/additional_experiments.py	
read	ch06/02_bonus_additional-experiments/gpt_download.py	
read	ch06/02_bonus_additional-experiments/previous_chapters.py	
read	ch06/03_bonus_imdb-classification/download_prepare_dataset.py	
read	ch06/03_bonus_imdb-classification/gpt_download.py	
read	ch06/03_bonus_imdb-classification/previous_chapters.py	
read	ch06/03_bonus_imdb-classification/sklearn-baseline.ipynb	
read	ch06/03_bonus_imdb-classification/train_bert_hf.py	
read	ch06/03_bonus_imdb-classification/train_bert_hf_spam.py	
read	ch06/03_bonus_imdb-classification/train_gpt.py	
read	ch06/03_bonus_imdb-classification/train_gpt_muon.py	
read	ch06/03_bonus_imdb-classification/train_sklearn_logreg.py	
read	ch06/04_user_interface/app.py	
read	ch07/01_main-chapter-code/ch07.ipynb	
read	ch07/01_main-chapter-code/exercise-solutions.ipynb	
read	ch07/01_main-chapter-code/exercise_experiments.py	
read	ch07/01_main-chapter-code/gpt_download.py	
read	ch07/01_main-chapter-code/gpt_instruction_finetuning.py	
read	ch07/01_main-chapter-code/instruction-data-with-response.json	
read	ch07/01_main-chapter-code/instruction-data.json	
read	ch07/01_main-chapter-code/load-finetuned-model.ipynb	
read	ch07/01_main-chapter-code/ollama_evaluate.py	
read	ch07/01_main-chapter-code/previous_chapters.py	
read	ch07/01_main-chapter-code/tests.py	
read	ch07/02_dataset-utilities/config.json	
read	ch07/02_dataset-utilities/create-passive-voice-entries.ipynb	
read	ch07/02_dataset-utilities/find-near-duplicates.py	
read	ch07/02_dataset-utilities/instruction-examples-modified.json	
read	ch07/02_dataset-utilities/instruction-examples.json	

Event	Path or command	Hash
read	ch07/03_model-evaluation/config.json	
read	ch07/03_model-evaluation/eval-example-data.json	
read	ch07/03_model-evaluation/llm-instruction-eval-ollama.ipynb	
read	ch07/03_model-evaluation/llm-instruction-eval-openai.ipynb	
read	ch07/03_model-evaluation/scores/correlation-analysis.ipynb	
read	ch07/03_model-evaluation/scores/gpt4-model-1-response.json	
read	ch07/03_model-evaluation/scores/gpt4-model-2-response.json	
read	ch07/03_model-evaluation/scores/llama3-8b-model-1-response.json	
read	ch07/03_model-evaluation/scores/llama3-8b-model-2-response.json	
read	ch07/04_preference-tuning-with-dpo/create-preference-data-ollama.ipynb	
read	ch07/04_preference-tuning-with-dpo/dpo-from-scratch.ipynb	
read	ch07/04_preference-tuning-with-dpo/instruction-data-with-preference.json	
read	ch07/04_preference-tuning-with-dpo/previous_chapters.py	
read	ch07/05_dataset-generation/config.json	
read	ch07/05_dataset-generation/instruction-data-llama3-7b.json	
read	ch07/05_dataset-generation/llama3-ollama.ipynb	
read	ch07/05_dataset-generation/reflection-gpt4.ipynb	
read	ch07/06_user_interface/app.py	
read	conftest.py	
read	pixi.toml	
read	pkg/llms_from_scratch/__init__.py	
read	pkg/llms_from_scratch/appendix_a.py	
read	pkg/llms_from_scratch/appendix_d.py	
read	pkg/llms_from_scratch/appendix_e.py	
read	pkg/llms_from_scratch/ch02.py	
read	pkg/llms_from_scratch/ch03.py	
read	pkg/llms_from_scratch/ch04.py	
read	pkg/llms_from_scratch/ch05.py	
read	pkg/llms_from_scratch/ch06.py	
read	pkg/llms_from_scratch/ch07.py	
read	pkg/llms_from_scratch/generate.py	
read	pkg/llms_from_scratch/kv_cache/__init__.py	
read	pkg/llms_from_scratch/kv_cache/generate.py	
read	pkg/llms_from_scratch/kv_cache/gpt2.py	
read	pkg/llms_from_scratch/kv_cache/llama3.py	
read	pkg/llms_from_scratch/kv_cache/qwen3.py	
read	pkg/llms_from_scratch/kv_cache/utils.py	
read	pkg/llms_from_scratch/kv_cache_batched/__init__.py	
read	pkg/llms_from_scratch/kv_cache_batched/generate.py	
read	pkg/llms_from_scratch/kv_cache_batched/qwen3.py	
read	pkg/llms_from_scratch/kv_cache_batched/utils.py	
read	pkg/llms_from_scratch/llama3.py	
read	pkg/llms_from_scratch/qwen3.py	
read	pkg/llms_from_scratch/tests/test_appendix_a.py	
read	pkg/llms_from_scratch/tests/test_appendix_d.py	
read	pkg/llms_from_scratch/tests/test_appendix_e.py	
read	pkg/llms_from_scratch/tests/test_ch02.py	
read	pkg/llms_from_scratch/tests/test_ch03.py	
read	pkg/llms_from_scratch/tests/test_ch04.py	
read	pkg/llms_from_scratch/tests/test_ch05.py	

Event	Path or command	Hash
read	pkg/llms_from_scratch/tests/test_ch06.py	
read	pkg/llms_from_scratch/tests/test_ch07.py	
read	pkg/llms_from_scratch/tests/test_generate.py	
read	pkg/llms_from_scratch/tests/test_kv_cache_context_limits.py	
read	pkg/llms_from_scratch/tests/test_kv_cache_gpt2.py	
read	pkg/llms_from_scratch/tests/test_llama3.py	
read	pkg/llms_from_scratch/tests/test_qwen3.py	
read	pkg/llms_from_scratch/utils.py	
read	pyproject.toml	
read	setup/.vscode/extensions.json	
read	setup/02_installing-python-libraries/python_environment_check.ipynb	
read	setup/02_installing-python-libraries/python_environment_check.py	
read	setup/02_installing-python-libraries/tests.py	
read	setup/03_optional-docker-environment/.devcontainer/devcontainer.json	
read	setup/04_optional-aws-sagemaker-notebook/cloudformation-template.yml	
read	ch06/01_main-chapter-code/sms_spam_collection/SMSSpamCollection.tsv	
write	ch06/01_main-chapter-code/train.csv	c89a1156b8476e6f
write	ch06/01_main-chapter-code/validation.csv	0ac7a07215729476
write	ch06/01_main-chapter-code/test.csv	a5764d4eed041dad
read	ch06/01_main-chapter-code/gpt2/124M/checkpoint	
read	ch06/01_main-chapter-code/gpt2/124M/model.ckpt.index	
read	ch06/01_main-chapter-code/gpt2/124M/model.ckpt.data-00000-of-00001	
write	ch06/01_main-chapter-code/loss-plot.pdf	4a5372b8a1c6d0ca
write	ch06/01_main-chapter-code/accuracy-plot.pdf	501616d1638a57e5
process	python -u gpt_class_finetune.py	
process	bash kv_run.sh	
process	tee ../../finetune.out	

C Run history

Invocation 1, OK, 28.5 seconds, 40 output lines, started 2026-10-07T15:54:12

metric_hash a35517aec5ff89c21ea92338af0bcfc6d5297c035d993e295fcbdb0f4977d5a1c
run_digest 4b465107063927f8985998fb53408dc93fddb633623e6a62736a1323efa30c0e
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 0e679d2750be501506cdce3c313cf4c7517b8764e6495f6425cfc40170929ce8
signed_message_hash d55ae7bdfdb53d9ef5cacb19f85e33c4a3f5c1ab8d79d0d3da44be46a8a049bf
nonce 374676608d623ff33d22f200227fdf1d
signed_at 2026-10-07T15:54:41.642621+00:00
sealed_at 2026-10-07T15:54:41.699138429Z
checkout_bundle c48ba9b0e18a012a4ec84ffd175e42d62e13f1ecf9979d7a4a2a5631760909c4

Signature (base64)

ovt0thSge9waew5nGNrPpo9fvBubeZ09NS3X8s/pXmus0L907pFkqv6L1H28Ahu8lP4zh2Lp35fUmWmmVaUKvvyf
tnGR7N52lkr4IRdqVuC2/20ysNz0U0RwUPEmdDHkZPHcHm62YruGGjQjo2MSzZJUjdtvHC/QyD1nFSkmvUvu/e5
FB72k/9g3SMJjhzy7SR06Wd8w4VLlYTFy+YVDyoxL8Gv90eEFSCs1SfLWaf0wPpqCYuYmH0hyrq1mNld4Xv2KA
KvPEj15WYq7z5gtL8/SPg7SiSa1rs0SRUjT0SLx1003oVld103QENhAl1yDYwt8p4g2Png0BitTvDbIN4rQvUEYu
1GHPNhhqJ3yFKOH5IDA1SEDZsR8KMDqw1u8ie1Godyd9zw6iQvVKsTJaSfIncvGHS5hRqW0nkYeXfYymrYPmLHS
FU0vr0fT64w7XfeibsdDJV9PSaEk0QLuf/Kg/+4s8YQixy5iMvVfpCsjMBTKkrGs32paE2sUctAp5gh4pCDh09tC

aqxNIwlqt5L5701Ur08JJbv0moGrU8rhh0hVxIWF8VX3+68a8Jz2Sd0qOrVTPYwKx8v5b0RkqEe02wDt+4Q+tKfR
1yUGNqWMVX60Nu0yPU7NYoMBjPMsjdVEWOK1JIhTrepU86hBDTmakxt5lrm8pqRaIrU=

Public key

-----BEGIN PUBLIC KEY-----

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sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4Lat4vcgWP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNPpxYa5X
7qU95EfEisX/GfoS9ToXtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KIcEadbIbHN
av0JjdTJ//CMOWZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWEKjwD5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqeS7VSZt3o
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6HEMLdkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7LE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCkDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRMb8LPP9v
8Jap/1PWjncNku5cdw3gHLCCAWEAAQ==

-----END PUBLIC KEY-----

Run 1 outputs

```
stdout c990e12ff9822769743352708a8377363351f18aa313ba24fa4372b8dba3a390
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics a35517aec5ff89c21ea92338af0bcfc6d5297c035d993e295fcbd0f4977d5a1c
digest 4b465107063927f8985998fb53408dc93fddb633623e6a62736a1323efa30c0e
kv_run.sh (unchanged)
pre aff3e36c2c15f779657806173c850e2ec420d8685a5d87d8ce1ebed06dee2fcc
post aff3e36c2c15f779657806173c850e2ec420d8685a5d87d8ce1ebed06dee2fcc
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```